NANDAN VARMA PERICHARLA

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SUMMARY

Systems engineer who builds things from scratch: OS kernels in C and Rust, including a capability-based microkernel built for virtualized, cloud-native workloads. Comfortable working in freestanding C, Rust `no_std`, UEFI firmware, and x86_64/ARM64.

EDUCATION

Seattle University - M.S. Computer Science | GPA: 3.5/4.0 **June 2026**

Relevant Coursework: Distributed Systems, Software Architecture and Design, Software Engineering Fundamentals

JNTU BVRIT - B.Tech. Computer Science & Data Science | GPA: 8.23/10 (≈3.7/4.0) **Aug 2024**

Relevant Coursework: Fundamentals of Operating Systems, Linux Programming, Computer Networks, Advanced Data Structures

PROFESSIONAL EXPERIENCE

Software Engineer Intern | Highspot Seattle, WA | **Jul 2025 – Sep 2025**

- Designed and shipped an AI feature using LLM tool-calling across multiple microservices, cutting user intent-to-action time by **50%** and reaching **20% adoption** in the first week.
- Engineered an LLM-powered dynamic compilation engine that tracked content changes over time and automatically regenerated accurate outputs, reducing stale content issues.
- Built an intelligent meeting summarization tool that searched distributed content repositories and injected relevant video clips into summaries for visual context.
- Built a React chat interface and inline content summarization features deployed to production, reducing time users spent searching large content collections.

TECHNICAL SKILLS

Languages: C, C++, Rust, Go, Swift, Python, Java, TypeScript, Bash
Systems: OS kernels, UEFI, x86_64, ARM64, QEMU, capability-based security, IPC, FAT VFS
Cloud & DevOps: Linux, Docker, CI/CD, AWS, GCP, Azure
Distributed Systems: Microservices, fault-tolerant service design, scalable data pipelines
Testing & Automation: Unit and contract tests, regression testing, host-test harnesses, CI + release pipelines

PROJECTS

Sigrun - Capability-Based Microkernel nandan-varma/sigrun **2026**

Rust (no_std), x86_64, GRUB2 multiboot2, QEMU, Cargo workspaces

- Booted a Rust microkernel end-to-end in QEMU (GRUB2 multiboot2, long-mode transition, APIC timer, round-robin scheduler) and ran a ring-3 init process over a hand-written SYSCALL/SYSRET path.
- Designed an seL4-style object-capability security model - unforgeable capability references, per-process capability tables, capability-gated IPC - targeting virtualized cloud workloads (KVM, Xen).

MarsOS - Desktop Operating System nandan-varma/mars nandan.fyi/projects/mars **2026**

C (freestanding), UEFI/OVMF, QEMU, Docker, Make

- Built a UEFI-booted x86_64 OS from scratch in freestanding C with no libc: cooperative multitasking kernel, capability-based process model, syscall interface, and virtual memory with page and heap allocators.
- Wrote a FAT VFS for persistent storage, plus a host-test harness that runs contract tests on the scheduler, allocators, event bus, and VM before every QEMU boot.

Pluto OS - Multi-Architecture Operating System nandan-varma/pluto **2025–2026**

Rust (no_std), x86_64, ARM64, RISC-V64, QEMU, Cargo workspaces

- Building a modular Rust (no_std) OS bootable on three architectures - x86_64, ARM64, RISC-V64 - via QEMU, with a Hardware Abstraction Layer keeping kernel and drivers portable from a single Cargo workspace.
- Structured as a Cargo workspace of independently compilable members: kernel, arch-specific bootloader, drivers, and an interactive shell.

Textedit - GPU-Accelerated Text Editor nandan-varma/textedit **2025–2026**

Rust, wgpu (WebGPU), winit, ropey, fontdue, syntect, GitHub Actions

- Built a text editor from scratch in Rust with a hand-rolled wgpu GPU rendering pipeline: custom glyph cache, text layout engine, cursor, and scrollbar - no GUI framework, all primitives rendered directly to the GPU.
- Wired up a dead test suite that had never actually run, surfacing real bugs that had shipped undetected, then shipped v0.1.1 through a CI + release pipeline for Linux and macOS.

LEADERSHIP & ACTIVITIES

Technical Coordinator, Metarics Data Science Club, BVRIT (2021–2024): built club websites, raised event participation by **30%**, led evaluation across 4 hackathons reviewing 150+ projects.

Team Lead, Smart India Hackathon (2022–2023): led a 6-member team to ship a women's safety emergency app under strict deadlines.